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Research Statement

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Should the next broadband dollar build a network or lower a bill? I study that question, and its analogues in Senegalese fintech markets and American bail courts, with structural models in markets where entry is costly and incumbents are entrenched. Where the standard toolkit cannot handle the setting, I build the piece it is missing: a moment-inequality bound where the entry game is too large to solve, a diagnostic where a causal-inference design leans on an assumption the data cannot support.

My job market paper, **"Build or Subsidize? The Welfare Effects of Deployment and Affordability Subsidies in U.S. Broadband,"** studies the two largest recent federal broadband programs: the $14.2 billion Affordable Connectivity Program (ACP), a demand-side subsidy, and the $42.45 billion Broadband Equity, Access, and Deployment program (BEAD), a supply-side subsidy. I combine cross-state event studies with a tract-level structural model of demand, pricing, and entry fit to 170,419 products across 66,454 census tracts. The entry game is too large to solve exactly (727,601 potential provider-tier positions), so I bound provider fixed costs with a moment-inequality estimator (Fan and Yang, 2024, JPE): it requires only that the market configuration I observe beat the alternatives a provider could have chosen, not a solved equilibrium or an assumed rule for picking among them.

The event studies show ACP lowered prices 7 to 10 percent at peak and shifted demand toward faster plans. BEAD's effects are smaller so far and lag its multi-year construction clock, consistent with a program still mid-rollout, not with a weak one. The structural model turns these patterns into welfare: BEAD delivers at least $12, and about $26 under my baseline equilibrium-selection rule, per subsidy dollar over 25 years, while ACP's measured gains roughly offset its outlays. The two programs read as complements, not substitutes: BEAD's gains concentrate in unserved and monopoly markets and are nearly flat across income; ACP's gains concentrate among low-income households and are nearly flat across market structure. Together they suggest a sequencing logic for future broadband spending: build where deployment is uneconomic, then subsidize affordability where price still binds. The paper was a finalist for the 2025 Bank of Canada Graduate Student Paper Award and was presented, by invitation, at the Competition Bureau of Canada.

The same tension between a strategic barrier and a structural one shows up in a very different market. In **"Fintech Entry and Agent Selection in Mobile Money Markets: Evidence from Senegal,"** I study Wave's expansion against telecom incumbents Orange, Free, and Expresso, using a retail audit of 1,550 stores and a moment-inequality entry game (Ciliberto and Tamer, 2009, Econometrica) that leaves equilibrium selection unrestricted. Wave reaches 14 percent of stores overall, but 76 percent of dedicated money-transfer outlets and 55 percent of stores whose operators own smartphones, against 3 percent of those without. The structural estimates trace this pattern to payoffs, not deterrence: smartphone readiness raises Wave's entry payoff by about 1.7, several times the roughly 0.2 effect of incumbent presence, and the estimates rule out zero for the digital-capacity parameters but not for any fintech-incumbent competition parameter. In counterfactuals, removing Wave raises incumbent coverage by less than one percentage point, while universal smartphone readiness more than doubles Wave's coverage with negligible incumbent displacement. As in unserved American counties, the binding constraint here looks like capacity, not competition.

Both applications pushed me past the tools available. The Senegal entry game exposed a specific flaw in how semiparametric matching handles selection inside moment-inequality models: when the actions we observe depend on structural errors the players themselves know, matching cannot remove the resulting bias, because the same payoff index that identifies the parameter also determines who enters. In **"Selection on Structural Errors in Moment-Inequality Estimation of Discrete Games,"** with Mohamed Coulibaly, we show that Ahn–Powell differencing (Ahn and Powell, 1993, J. Econometrics) still works for post-entry outcomes once the propensity score gives way to the vector of players' choice propensities, and we derive an envelope bound on the remaining selection term that holds under any rule for selecting among equilibria. In simulations and in the Senegal data, that bound turns an empty identified set into a nonempty, economically interpretable one.

A separate paper applies the same habit elsewhere in applied econometrics. Researchers increasingly plug machine-learning estimates of first-stage choice probabilities and state transitions into dynamic discrete choice models, a shortcut that can bias the resulting parameters and break standard inference. **"Efficient and Debiased Estimation of Dynamic Discrete Choice Models"** develops Neyman-orthogonal, debiased estimators (Chernozhukov et al., 2018, Econometrics J.) that stay root-$n$ consistent and asymptotically normal even when those first-stage objects come from machine learning.

My econometric agenda also runs through causal inference: a widely used empirical strategy deserves scrutiny before its estimates guide policy. Judge-IV designs, used to estimate the effects of incarceration, pretrial detention, and disability insurance, increasingly invoke average monotonicity (Frandsen, Lefgren, and Leslie, 2023, AEJ: Applied) in place of the harder-to-defend strong monotonicity assumption. In **"Target-Population Fragility in Judge IV under Average Monotonicity,"** with Mohamed Coulibaly, we show that average monotonicity fixes the sign of the IV weights but not the population they average over: the admissible response types switch discretely when a judge's propensity crosses the assignment-weighted average, so an arbitrarily small design change can shift the estimand to a different population. We propose a standardized boundary diagnostic and show that three of eight judges in the Philadelphia bail data are boundary-fragile, evidence that target-population stability belongs alongside the standard first-stage and exclusion checks.

These five papers do not share one estimator: entry games, dynamic discrete choice, and judge IV call for different tools. They share a habit: find the assumption a policy question or an empirical strategy is quietly leaning on, and when the setting does not justify it, build or repair the tool that relaxes it. Three of the five, the broadband and Senegal applications and the selection-bias paper, form one line of work in moment-inequality games; the other two extend the same habit into semiparametric estimation and causal inference. Where the next application needs a tool that does not yet exist, I intend to build it.